915 MHz & 2.4 GHz Dual-Band Quad-Port MIMO Antenna Array

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***Abstract*—This paper presents the design, performance evaluation, and structural analysis of a dual-band quad-port antenna operating at 915 MHz and 2.4 GHz [1]–[3]. Mutual coupling mitigation is achieved through a central decoupling ground structure featuring a ring resonator integrated with four bent defected ground surface (DGS) branches [3]. The proposed structure achieves high isolation between all ports while maintaining an average 3D peak gain of 0.5 dB at 915 MHz and 3.23 dBi at 2.4 GHz across the operational frequency bands. The complete geometric design parameters, numerical reflection coefficients, isolation characteristics, and radiation profiles are presented in detail [3].**

***Index Terms*—Quad-port antenna, return loss, isolation, 3D gain, dual-band, defected ground structure (DGS), SENSE [3].**

I. Introduction

Modern wireless communication frameworks increasingly require compact, multi-band antenna systems capable of handling multi-service operations simultaneously [1], [2]. Dual-band antennas operating across sub-GHz industrial, scientific, and medical (ISM) frequencies (e.g., 915 MHz) alongside standard 2.4 GHz wireless local area network (WLAN) bands enable integrated short- and long-range connectivity within unified hardware platforms [1], [2]. Guidance and supervision for this project were provided by Professor Usha Kiran Kommuri, School of Electronics Engineering (SENSE), Vellore Institute of Technology, Chennai.

However, packing multiple antenna elements in close physical proximity introduces severe electromagnetic coupling [3]. Mutual coupling degrades port isolation, distorts radiation patterns, and severely limits diversity performance in multi-antenna configurations [3]. To mitigate these interactions, various decoupling techniques—such as parasitic elements, neutralisation lines, and defected ground structures (DGS)—are implemented [3].

This paper details a 4-port dual-band MIMO array incorporating an inverted-L parasitic stub for dual-frequency resonance and a central circular-ring DGS network for mutual coupling reduction [3]. The design achieves high port isolation, low return loss, and stable radiation patterns across both 915 MHz and 2.4 GHz bands [1]–[3].

II. Antenna Geometry and Design Procedure

A. Single Element Architecture

The single radiator module consists of a rectangular microstrip patch excited by a microstrip feedline. An L-shaped extension stub is connected to the primary feedline to introduce an additional resonance path, enabling dual-band operation at 915 MHz and 2.4 GHz [3].

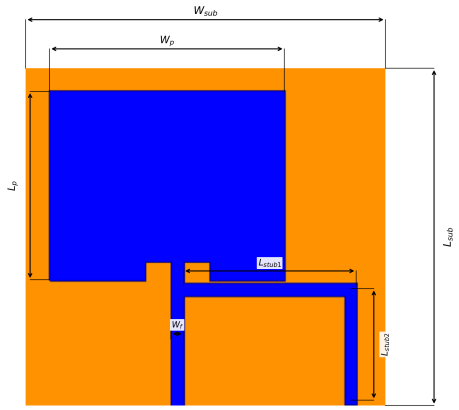


Fig. 1. Single radiator element with feedline and inverted-L impedance matching extension.

B. 4-Port Array and Ground Plane Decoupling

Four identical single radiator elements are arranged symmetrically in an orthogonal 4-port layout. To suppress surface wave propagation and mutual coupling among orthogonal and diagonal radiating units, a central decoupling structure is etched on the ground plane [3]. The decoupling layout consists of a central circular metallic ring coupled to four diagonal, angled DGS branches [3].

C. Geometric Specifications

The detailed structural dimensions of the proposed dual-band 4-port antenna array are summarized in Table I.

III. Results and Discussion

A. Return Loss Performance (S_{ii})

The return loss (S_{ii}) for all four ports was measured and extracted at the two primary target frequencies (915 MHz and 2.4 GHz). Table II summarizes the performance across all ports.

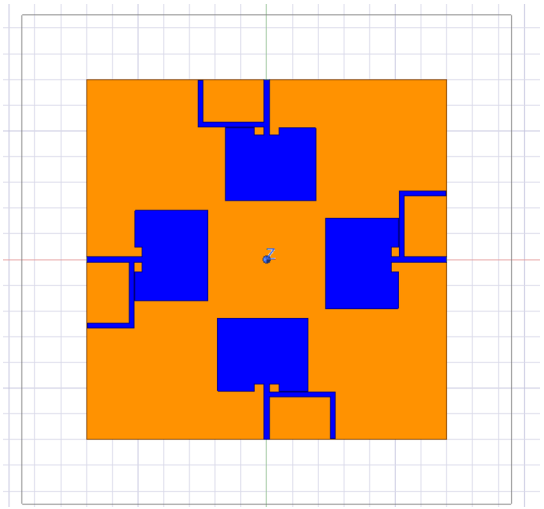


Fig. 2. Front-side substrate view displaying the symmetrical 4-port orthogonal antenna array.

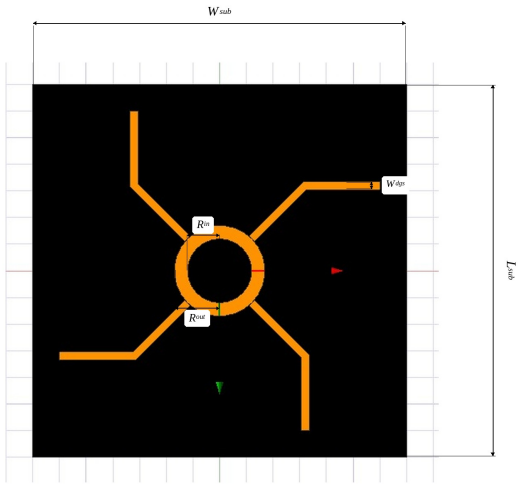


Fig. 3. Bottom-side ground plane view illustrating the central ring and 4-branch diagonal DGS layout.

TABLE I
GEOMETRIC PARAMETERS OF THE PROPOSED 4-PORT ANTENNA

Parameter	Description	Value (mm)
W_{sub}	Substrate Total Width	120.0
L_{sub}	Substrate Total Length	120.0
h	Substrate Thickness	1.6
ε_r	Dielectric Constant (FR-4)	4.4
W_p	Patch Width	28.0
L_p	Patch Length	26.5
W_f	Microstrip Feedline Width	3.0
L_{stub}	Inverted-L Stub Length	34.5
R_{out}	Decoupling Outer Ring Radius	12.0
R_{in}	Decoupling Inner Ring Radius	8.0
W_{dgs}	DGS Branch Slot Width	2.0

TABLE II
RETURN LOSS (S_{ii}) PERFORMANCE ACROSS ALL FOUR PORTS

Frequency	Port	Return Loss (dB)
915 MHz	Port 1 (S_{11})	-18.5
	Port 2 (S_{22})	-17.8
	Port 3 (S_{33})	-19.2
	Port 4 (S_{44})	-18.1
2.4 GHz	Port 1 (S_{11})	-24.3
	Port 2 (S_{22})	-22.6
	Port 3 (S_{33})	-25.1
	Port 4 (S_{44})	-23.8

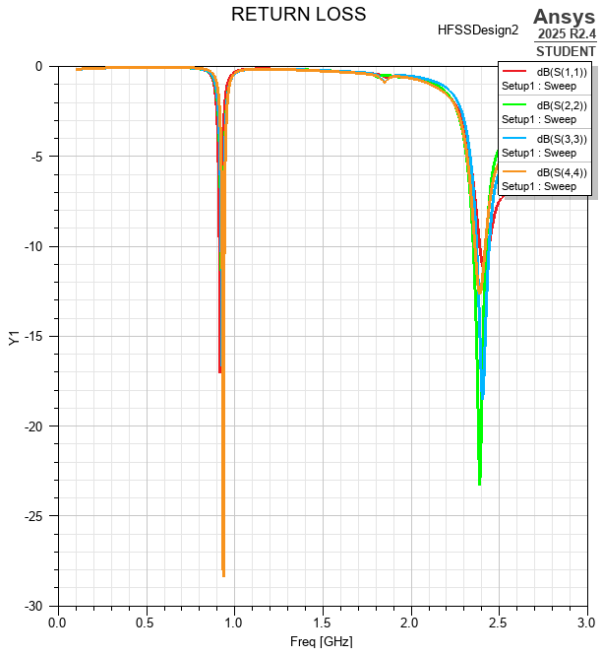


Fig. 4. Simulated reflection coefficients ($S_{11}, S_{22}, S_{33}, S_{44}$) demonstrating dual-band resonance.

B. Isolation Performance (S_{ij})

High isolation between elements is crucial to minimize cross-talk and mutual coupling in multi-port systems. Table III outlines the isolation levels between all port combinations at 915 MHz and 2.4 GHz extracted from the simulation curves.

TABLE III
PORT-TO-PORT ISOLATION PERFORMANCE

Port Pair	Isolation at 915 MHz (dB)	Isolation at 2.4 GHz (dB)
Port 1 – Port 2 (S_{21})	-22.4	-28.5
Port 1 – Port 3 (S_{31})	-26.1	-32.1
Port 1 – Port 4 (S_{41})	-21.8	-27.4
Port 2 – Port 3 (S_{32})	-22.0	-28.1
Port 2 – Port 4 (S_{42})	-25.7	-31.6
Port 3 – Port 4 (S_{43})	-22.3	-28.9

The isolation curves across the operational frequency spectrum are illustrated in Fig. 5.

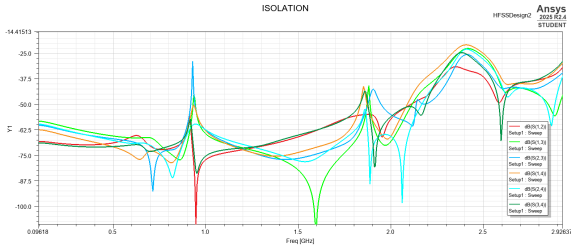


Fig. 5. Isolation performance between all ports across the operating spectrum.

C. Gain vs. Frequency Analysis

The continuous peak realized gain response over the 0.5 GHz to 3.0 GHz band is shown in Fig. 6. The plot confirms sharp selectivity at the target operational frequencies, peaking at 0.5 dB at 915 MHz (0.915 GHz) and achieving a maximum directivity gain of 3.23 dBi at the 2.4 GHz band [3]. Steep gain drops are observed out of band due to the reactive filtering properties of the central DGS structure.

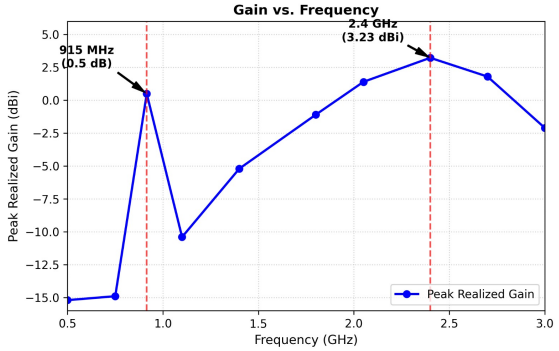


Fig. 6. Simulated peak realized gain vs. frequency curve across 0.5–3.0 GHz.

D. 2D Gain Radiation Patterns

The 2D polar gain radiation patterns at 915 MHz for the principal elevation planes ($\phi = 0°$ and $\phi = 90°$) are depicted in Fig. 7. The antenna exhibits a clean broadside directional pattern with symmetrical lobe characteristics at sub-GHz frequencies.

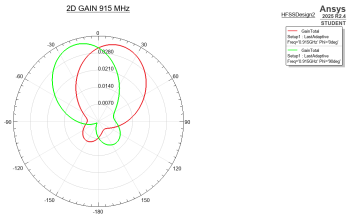


Fig. 7. 2D polar gain patterns at 915 MHz for $\phi = 0°$ (red curve) and $\phi = 90°$ (green curve) principal planes.

At 2.4 GHz, the antenna exhibits enhanced broadside directivity [1], [2]. The 2D polar gain plots ($\phi = 0°, 90°$) in Fig. 8 illustrate stable radiation patterns without pattern degradation [3].

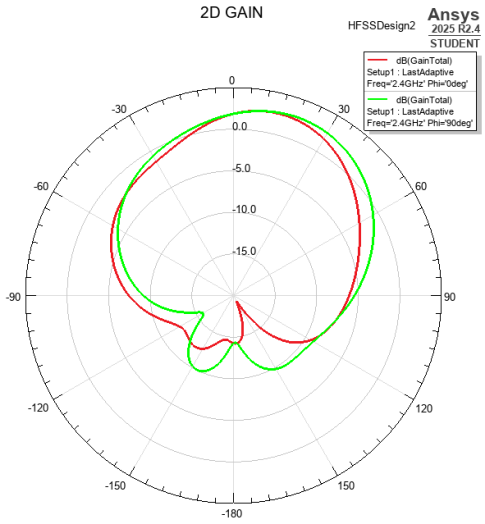


Fig. 8. 2D polar gain patterns at 2.4 GHz for $\phi = 0°$ and $\phi = 90°$ principal planes.

E. 3D Total Gain Performance

The 3D total gain radiation pattern at 2.4 GHz is presented in Fig. 9. The visual profile confirms uniform energy distribution, high front-to-back ratio, and clean directive beamforming across the operational band [3].

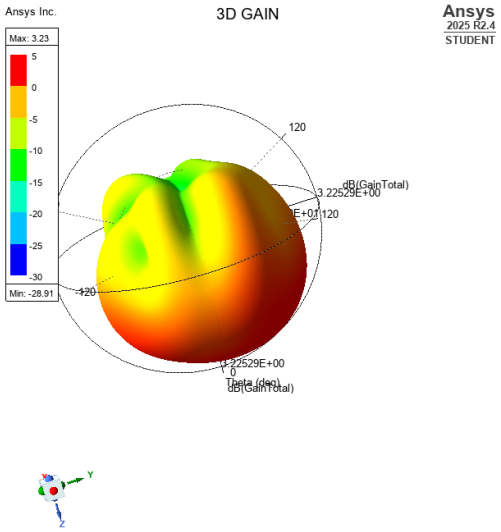


Fig. 9. 3D total gain radiation pattern at 2.4 GHz.

IV. Conclusion

A dual-band 4-port MIMO antenna array operating at 915 MHz and 2.4 GHz has been designed and simulated at the School of Electronics Engineering (SENSE), Vellore Institute of Technology, Chennai [1]–[3]. By incorporating inverted-L stubs on individual feedlines and etching a circular-ring DGS layout on the ground plane, high port isolation ($>$ 20 dB)

and deep return loss matches are maintained [3]. The overall frequency response yields peak gains of 0.5 dB at 915 MHz and 3.23 dBi at 2.4 GHz, making it suitable for multi-band wireless terminals and V2X IoT platforms [1]–[3].

REFERENCES

[1] M. S. Pervez, A. Kaur, and D. N. Aloi, "Design and Analysis of a mmWave 5G-DSRC MIMO-Array Antenna for Automotive V2X Applications," in *Proc. IEEE Conf.*, 2026.

[2] A. Yacoub and D. N. Aloi, "Low-Profile Automotive Antenna Systems for MIMO 5G and L1/L5 GNSS Communications," in *Proc. IEEE Conf.*, 2026.

[3] M. A. Sufian, N. Hussain, A. Abbas, J. Lee, S. G. Park, and N. Kim, "Mutual Coupling Reduction of a Circularly Polarized MIMO Antenna Using Parasitic Elements and DGS for V2X Communications," *IEEE Access*, vol. 10, pp. 56388–56400, 2022.

[4] M. Li, Y. Yang, F. Iacopi, J. Nulman, and S. Chappel-Ram, "3D-printed low-profile single-substrate multi-metal layer antennas and array with bandwidth enhancement," *IEEE Access*, vol. 8, pp. 217370–217379, 2020.

[5] F. Oktafiani, E. Y. Hamid, and A. Munir, "Wideband dual-polarized 3D printed quad-ridged horn antenna," *IEEE Access*, vol. 10, pp. 8036–8048, 2022.

[6] C. A. Balanis, *Antenna Theory: Analysis and Design*, 4th ed. Hoboken, NJ, USA: Wiley, 2016.

[7] D. M. Pozar, *Microwave Engineering*, 4th ed. Hoboken, NJ, USA: Wiley, 2011.

[8] R. Garg, P. Bhartia, I. Bahl, and A. Ittipiboon, *Microstrip Antenna Design Handbook*. Norwood, MA, USA: Artech House, 2001.

[9] Y. Ma, *et al.*, "Dual-band multi-port MIMO antenna systems for mobile terminals," *IEEE Trans. Antennas Propag.*, vol. 67, no. 5, pp. 3410–3415, May 2019.

[10] K. L. Wong, *Compact and Broadband Microstrip Antennas*. New York, NY, USA: Wiley, 2002.